

Week 8 assignment is, inevitably, about Regular Expressions and OpenRefine

Upload a text file or a PDF with solutions to the problems below. Use the first page of this [template](#) if you like. Beware of making the submission legible and understandable to another reader; for example, consider using the "Save regex" functionality in [regex101.com](#) to create a link to your solution and add the link to your answer. Remember that you can elaborate solutions in groups, but need to submit **individually**.

Assignment 8.1 & 8.2

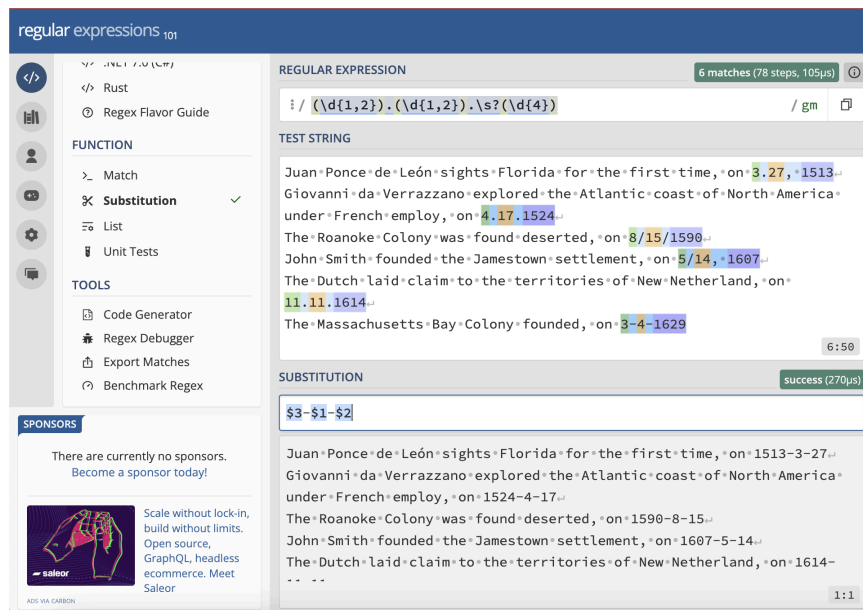
What regular expressions do you use to extract all the dates in this blurb:

<http://bit.ly/regexexercise2> and to put them into the following format YYYY-MM-DD ?

First step is plotting the text, found through the link above, into the test string. In the regular expression bar is typed the following expressions, to first and foremost find all the dates in the text, regardless of how the dates are written – meaning dd-MM-yy or d/M/yyyy etc.:

`(\d{1,2}).(\d{1,2}).\s?(\d{4})`

“\d”, matches any single digit, which means whatever expression follows, refers to any kind of digit. With that said, the next expression is “{1,2}”, which means the digits can be between one and two ciphers or numbers. This is followed by a “.”, which initially can refer to any kind of character between the numbers (referring to date, month and year) – could be /, -, space etc.. This expression is put within parentheses to single put this particular search, even though the expressions continue, referring to month and year, which equally so are encased in parentheses. At the end the expression “{4}”, like the expression “{1,2}”, simply refers to the search of a line of numbers that come in pairs of 4.



To change the dates into the following format **YYYY-MM-DD**, the following expressions are typed into the substitution bar:

\$3-\$1-\$2

Substitution: \$3 refers to yyyy, \$1 refers to MM, \$2 refers to dd. By plotting the expressions in, in that order, the original text in the test string will be converted into the correct format of **YYYY-MM-DD**, from whatever format it was listed as before.

Assignment 8.3

Write a regular expression to convert the stopwordlist (list of most frequent Danish words) from Voyant in <http://bit.ly/regexexercise3> into a neat stopword list for R (which comprises "words" separated by commas, such as <http://bit.ly/regexexercise4>). Then take the stopwordlist from R <http://bit.ly/regexexercise4> and convert it into a Voyant list (words on separate line without interpunction)

Link to stopwordlist in voyant (Regex101 URL): <https://regex101.com/r/jwxvqc/2>

Link to stopwordlist in R (Regex101 URL): <https://regex101.com/r/KS08tK/1>

Assignment 8.4

Does OpenRefine alter the raw data during sorting and filtering?

Yes. By, for instance, pin pointing misspelling in categories listed under the headliner, or separating the listed every cell, as shown below.

By clicking the small arrow on a top bar cell, then edit and finally transformation, the following expressions can be typed in:

```
value.replace("[", "").replace("]", "").replace(" ", "").replace("","")
```

Regular expressions are used to match and replace certain data/grammar. Separate the words from each other in the cells, by removing semikolon. By clicking the small arrow on a top bar cell, then facet, custom text facet and type:

```
value.split(";")
```

By then the data will have gone from 96 choices to now 18 choices, changing the data – because we made it more tidy:)

The screenshot shows the OpenRefine interface with a CSV file named 'SAFI openrefine.csv'. The main table has 131 rows. The 'Items_owned' column contains various items listed in a single cell, separated by semicolons. The left sidebar shows a facet for 'Items_owned' with 96 choices. Below it, a second facet shows the result of the transformation, with 18 choices. The choices are: bicycle (60), car (3), computer (2), cow_cart (30), cow_plough (85), electricity (6), fridge (5), lorry (5), mobile_phone (86), motorcycle (39), and NULL (10).

Interview_date	Items_owned
17-Nov-16	bicycle;television;solar_panel;table
17-Nov-16	cow_cart;bicycle;radio;cow_plough;solar_panel;solar_torch;table;mobile_phone
17-Nov-16	solar_torch
17-Nov-16	bicycle;radio;cow_plough;solar_panel;mobile_phone
17-Nov-16	motorcycle;radio;cow_plough;mobile_phone
17-Nov-16	NULL
17-Nov-16	motorcycle;cow_plough
16-Nov-16	motorcycle;bicycle;television;radio;cow_plough;solar_panel;solar_torch;table;fridge
16-Nov-16	television;solar_panel;solar_torch
16-Dec-16	cow_cart;motorcycle;bicycle;television;radio;cow_plough;solar_panel;solar_torch;table
21-Nov-16	radio;cow_plough
21-Nov-16	cow_cart;bicycle;radio;cow_plough;table
21-Nov-16	bicycle;radio;cow_plough;mobile_phone
21-Nov-16	bicycle;radio;cow_plough;solar_panel;table;mobile_phone
21-Nov-16	bicycle;radio;cow_plough;solar_panel;table
24-Nov-16	radio;cow_plough;solar_panel;solar_torch
21-Nov-16	mobile_phone

Fix the [interviews dataset](#) in OpenRefine enough to answer this question:

"Which two months are reported as the most water-deprived/dryest by the interviewed farmer households?"

OpenRefine

Andrea Kesmodel

SAFI openrefine.csv.csv

Permalink

Open...

Export

Help

Facet / Filter

Undo / Redo 2 / 2

86 matching rows (131 total)

Extensions Wikibase

Refresh

Reset all

Remove all

Show as: rows records

Show: 5 10 25 50 100 500 1000 rows

« first

< previous

1

- 130

next >

last »

months_no_water

change

15 choices Sort by: name count

Cluster

[Sept'; 'Oct'; 'Nov'] 21

[Aug'; 'Sept'; 'Oct'; 'Nov'] 14

[Sept'; 'Oct'] 14

[Aug'; 'Sept'; 'Oct'] 11

[Oct'; 'Nov'] 8

[Aug'; 'Sept'] 6

[Jan'; 'Dec'] 2

[Nov'; 'Dec'] 2

[Oct'] 2

[Apr'; 'May'; 'June'; 'July'; 'Aug'; 'Sept'; 'Oct'; 'Nov'] 1

[Feb'; 'Aug'; 'Sept'; 'Oct'; 'Nov'] 1

no_enough_water

change invert reset

3 choices Sort by: name count

Cluster

exclude

yes 86

NULL 39

no 6

Facet by choice counts

▼ All

☆

2.

17-Nov-16

[Aug'; 'Sept']

yes

cow_cart;bicycle;radio;cow_plough;solar_panel;solar_torch;table;mobile_phone

1

▼ interview_date

☆

7.

17-Nov-16

[Aug'; 'Sept'; 'Oct']

yes

motorcycle;cow_plough

7

▼ months_no_water

☆

8.

16-Nov-16

[Sept'; 'Oct']

yes

motorcycle;bicycle;television;radio;cow_plough;solar_panel;solar_torch;table;fridge

8

▼ no_enough_water

☆

9.

16-Nov-16

[Oct'; 'Nov']

yes

television;solar_panel;solar_torch

9

▼ items_owned

☆

10.

16-Dec-16

[Sept'; 'Oct'; 'Nov']

yes

cow_cart;motorcycle;bicycle;television;radio;cow_plough;solar_panel;solar_torch;table

10

▼ quest_n

☆

12.

21-Nov-16

[Aug'; 'Sept'; 'Oct'; 'Nov']

yes

cow_cart;bicycle;radio;cow_plough;table

12

☆

13.

21-Nov-16

[Oct']

yes

bicycle;radio;cow_plough;mobile_phone

13

☆

15.

21-Nov-16

[Sept'; 'Oct'; 'Nov']

yes

bicycle;radio;cow_plough;solar_panel;table

15

☆

21.

21-Nov-16

[Oct']

yes

NULL

21

☆

24.

21-Nov-16

[Sept'; 'Oct']

yes

radio;table;sofa_set;mobile_phone

24

☆

25.

21-Nov-16

[Aug'; 'Sept']

yes

cow_cart;motorcycle;television;radio;cow_plough;solar_panel;solar_torch;table;sofa_set;mobile_phone

25

☆

26.

21-Nov-16

[Oct'; 'Nov']

yes

radio;cow_plough;table;mobile_phone

26

☆

28.

21-Nov-16

[Aug'; 'Sept'; 'Oct'; 'Nov']

yes

NULL

28

☆

29.

21-Nov-16

[Sept'; 'Oct'; 'Nov']

yes

motorcycle;bicycle;radio;table;mobile_phone

29

☆

32.

21-Nov-16

[Sept'; 'Oct']

yes

cow_cart;motorcycle;radio;cow_plough;solar_panel;mobile_phone

32

☆

33.

21-Nov-16

[Aug'; 'Sept'; 'Oct']

yes

cow_cart;lorry;motorcycle;sterio;cow_plough;solar_panel;mobile_phone

33

☆

34.

17-Nov-16

[Oct'; 'Nov']

yes

television;radio;cow_plough;solar_panel;solar_torch;table;mobile_phone

34

September, Oktober, November:)

Assignment 8.5

Real-Data Challenge: **What are the 10 most frequent occupations "erhverv" among unmarried men or women of 20-30 years in 1801 Aarhus census dataset?** (hint: first select either men or women to shrink the dataset to a manageable size, then filter by age, and then use merging to cut the erhvervvariation ruthlessly.)

Start by clicking at the all cell → edit columns → re-order / remove columns. The theme bars shown at the top (koen/køn, giftnr/marriage, enavn/last name, alder, erhverv) since it will be the data I'll be working with and most relevant to answer this question. with "alder" I included only those who are between 20-30 years of age in facet/filter. Afterwards under "giftnr" I included only blank – meaning those who aren't married. Then Finally the "erhverv", which first needed to be de-clustered in the facet/filter. By clicking count, the "erhverv" mentioned most often appears at the top and so on – before writing down the 10 first, I carefully, manually, went through the professions, which the computer had not detected as cluster, and edited the title in each cell, to correspond with a single title, including all data referring to the same profession. The following shows the visual from OpenRefine:

OpenRefine census 1801 normalized csv [Permalink](#) Open... Export Help

Facet / Filter Undo / Redo 235 / 235 **761 matching records (44,523 total)** Extensions Wikibase

Refresh Reset all Remove all Show as: rows records Show: 5 10 25 50 100 500 1000 records « first < previous 1 - 5390 next > last »

	koen	giftnr	enavn	alder	erhverv	ft	sogn	amt	id	fnavn	loknr	lokalitet	bygning
34.	mand		Nielsen	22	soldat	1801	Alra	Århus	34	Søren	3	Alrøe Bye	
50.	mand		Jensen	21	matros	1801	Alra	Århus	50	Jørgen	5	Alrøe Bye	
109.	mand		Nielsen	27	matros	1801	Alra	Århus	109	Rasmus	13	Alrøe Bye	
211.	mand		Nielsen	23	Enrouleret Matros	1801	Alra	Århus	211	Rasmus	35	Alrøe Bye	
217.	mand		Rasmussen	24	soldat	1801	Astrup	Århus	5	Rasmus	1	Astrup	
242.	mand		Jensen	27	vanføer	1801	Astrup	Århus	30	Michel	4	Astrup	
340.	mand		Rasmussen	28	soldat	1801	Astrup	Århus	128	Niels	25	Asch	
349.	mand		Rasmussen	24	soldat	1801	Astrup	Århus	137	Peder	26	Asch	
469.	mand		Povlsen	26	soldat	1801	Astrup	Århus	257	Christen	53	Beggeskov	
532.	mand		Fransen	25	soldat	1801	Beder	Århus	4	Søren	1	Beder Bye	
560.	kvinde		Pedersdatter	30	Væver	1801	Beder	Århus	32	Maren	4	Beder Bye	
662.	mand		Jensen	27	vanføer	1801	Beder	Århus	134	Søren	24	Fulden Bye	
773.	kvinde		Christophers datter	20	Almisser	1801	Beder	Århus	245	Karen	48	Fulden Bye	
798.	mand		Laursen	26	soldat	1801	Beder	Århus	270	Rasmus	52	Floystrup Bye	
852.	mand		Olesen	30	Skræder	1801	Beder	Århus	324	Rasmus	65	Floystrup Bye	
913.	mand		Erichsen	24	soldat	1801	Bjerager	Århus	60	Peder	10	Dyngbye	
941.	mand		Olesen	27	soldat	1801	Bjerager	Århus	88	Christian	15	Dyngbye	
1129.	mand		Rasmussen	23	Tieneste	1801	Bjerager	Århus	276	Erich	56	Rørth Bye	
1135.	kvinde		Jacobsdatter	20	Tieneste	1801	Bjerager	Århus	282	Johanne	57	Rørth Bye	
1313.	mand		Pedersen	24	soldat	1801	Borum	Århus	24	Jens	3	Borum Bye	
1499.	mand		Foss	20	Væver	1801	Borum	Århus	210	Jens	37	Borum Bye	
1500.	kvinde		Pedersdatter	26	Væver	1801	Borum	Århus	211	Johanne	37	Borum Bye	
1547.	mand		Christensen	29	Væver	1801	Borum	Århus	258	Jens	51	Borum Bye	

Facet by choice counts

giftnr change invert reset
1 choice Sort by: name count Cluster
1 9 (blank) 761 exclude

erhverv change invert reset
202 choices Sort by: name count Cluster
soldat 423 exclude
Tieneste 128 exclude
Væver 81 exclude
vanføer 23 exclude
matros 19 exclude
Snedker 16 exclude
Skræder 15 exclude
Land Recrut 14 exclude
gårdskarl 13 exclude
Almisser 12 exclude
Bonde og Gaardbeboer 12

The most frequent occupations "erhverv" among unmarried men or women of 20-30 years in 1801 Aarhus census dataset is the following:

- 1) Soldat
- 2) Tieneste
- 3) Væver
- 4) Vanføer
- 5) Matros
- 6) Snedker
- 7) Skræder
- 8) Land Recrut
- 9) Gårdskarl
- 10) Almisser